

Pump



Syringe pump (Chemyx: Fusion 100)

3대 보유 중

Syringe (Norm-ject plastic)	Min flow rate (mL/min)	Max flow rate (mL/min)	Max Pressure	Inventory
1 mL	0.0001	3.2452	<3 bar	지속적으로 사용할 수 있을 만큼 재고 보유함.
2 (3) mL	0.0001	13.8227		
5 (6) mL	0.0001	22.8812		
10 (12) mL	0.0001	37.0935		
20 (24) mL	0.0001	58.7368		
Syringe (Hamilton-gas tight)	Min flow rate (mL/min)	Max flow rate (mL/min)	Max Pressure	Inventory
2.5 mL	0.0001	7.8258	<13.8 bar	1 ea
10 mL	0.0001	31.3462		2 ea

Pump



Dual syringe pump
(Hamilton PSD/4)

PSD/4 Type Specifications	Highest Resolution (steps/stroke)	Fastest/Slowes (time/stroke)	Size DxHxW(in)	Linear Force Capability(lbf)	Valve Torque Capability (in oz)	Weight (lbs)
High Lift Force	54,857	2.3 sec/114.3 min	4.2 x 5 x 1.75	40	25	2.6

Syringe (Norm-ject plastic)	Min flow rate (mL/min)	Max flow rate (mL/min)	Max Pressure	Inventory
1 mL	0.000875	26.1	<13.8 bar	2 ea
5 mL	0.0437	130		2 ea

Pump

< Peristaltic pump >



E-series
(pump 3개)



SF-10
(pump 1개)

Flow rate	Pressure	Inventory
0.02 – 10 mL/min (0.01 단위 조작 가능)	<10 bar (recommend <8 bar)	Pump 4개 사용 가능

- 별도 첨부 파일 확인 (Blue, Red tubing 보유 중 입니다.)

< HPLC pump >



R-series
(pump 2개)



R2C+ pump
(pump 2개)

Flow rate	Pressure	Inventory
0.01 – 10 mL/min (0.01 단위 조작 가능)	<42 bar	Pump 4개 사용 가능

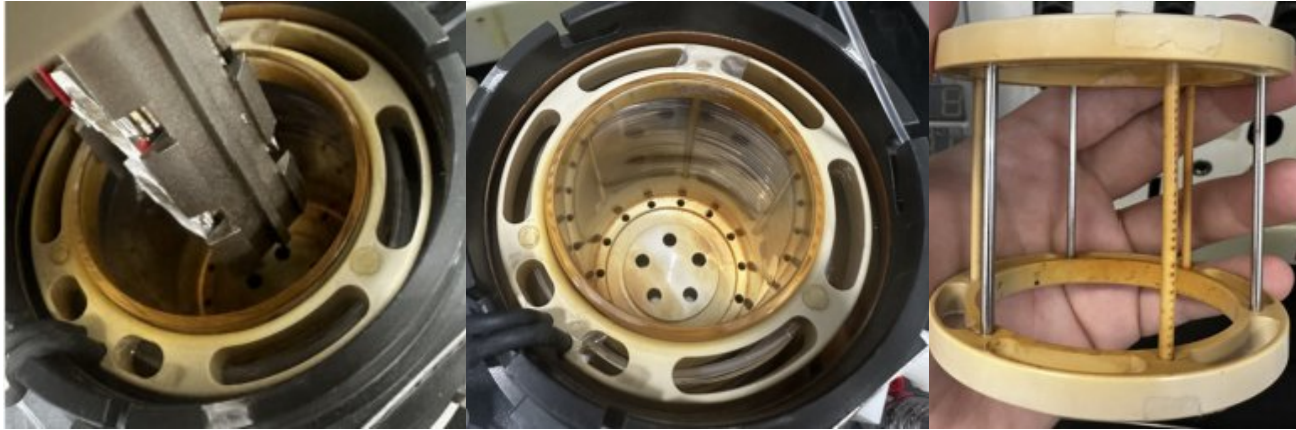
- 슬러리 / 현탁액 / 고체 입자가 섞인 시약 / 기체 (**사용 불가**)
- Resistance for acid: 질산, 황산, 염산, TFA 등 사용 가능
- **No resistance for Base: n-BuLi, LDA, LAH etc.**

Tubing reactor

Tubing ID	Volume (mL)	Material	Inventory	Pressure	Temperature
0.04" (1.0 mm)	2	PFA	1	<34 bar	80 °C
0.04" (1.0 mm)	5	PFA	2		80 °C
0.04" (1.0 mm)	10	PFA	1		80 °C
0.04" (1.0 mm)	10	FEP	1		50 °C
0.04" (1.0 mm)	20	FEP	1		50 °C
0.03" (0.75 mm)	5	PFA	1		80 °C

Photo reactor

UV-150 photo reactor



Wavelength (nm)	Input power	Radiant power	Wavelength (measured)	Optical power (mW/cm ²)	Temperature
365	60 W	3 W	-	-	35 – 80 °C (non-additional set-up) 10 – 35 °C (Additional setup is required.)
380	60 W	4.2 W	-	-	
395	60 W	6 W	396	-	
420	60 W	18 W	424	243	
450	60 W	24 W	450	355	
470	60 W	24 W	-	-	
525	60 W	3 W	533	-	

Cartridge에 tubing을 감아서 사용하기 때문에
0.1 mm tubing 사용 시에 반응기 부피가 **최대 10 mL**로 제한됩니다.

Manual photo reactor



Wavelength (nm)	Input power	Radiant power	Wavelength (measured)	Optical power (mW/cm ²)	Temperature
Blue	24 V	-	448	62	Fan : 40 °C No fan : 50 °C
Red	24 V	-	-	-	-
Green	24 V	-	-	-	-

Manual의 경우 tubing을 감을 수 있는 cartridge의 사이즈가
크기때문에 반응기 부피 **최대 30 mL**로 제한됩니다.

MFC

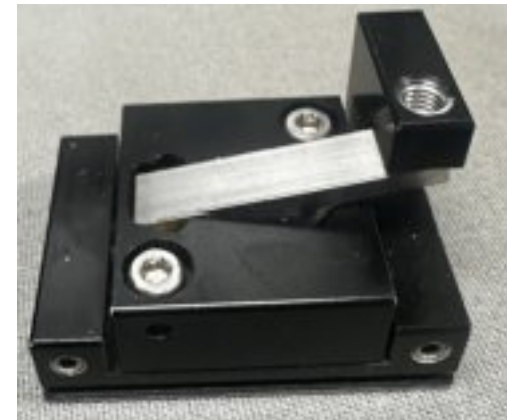


Gas	Inlet pressure	Outlet pressure	Flow range
O ₂	10 bar	9 bar	0.01 – 10 sccm (0.01 단위 조작 가능)

Pressure (bar)	Inventory
1.5	2 ea
2.5	1 ea
7.0	1 ea
8.0	2 ea
16 bar	1 ea
Adjustable	2 ea



Cartridge BPR



Adjustable BPR

Hydrogenation reactor



ThalesNano H-cube mini plus

Temperature range	25 – 100 °C
Pressure range	1 – 100 bar
Flow rate	0.1 – 3.0 mL/min
Hydrogen Generation	25 – 30 mL/min
Water Reservoir Capacity	100 mL
Catalyst Amount	0.1 – 0.3 g
Concentration	0.01 – 1.0 M

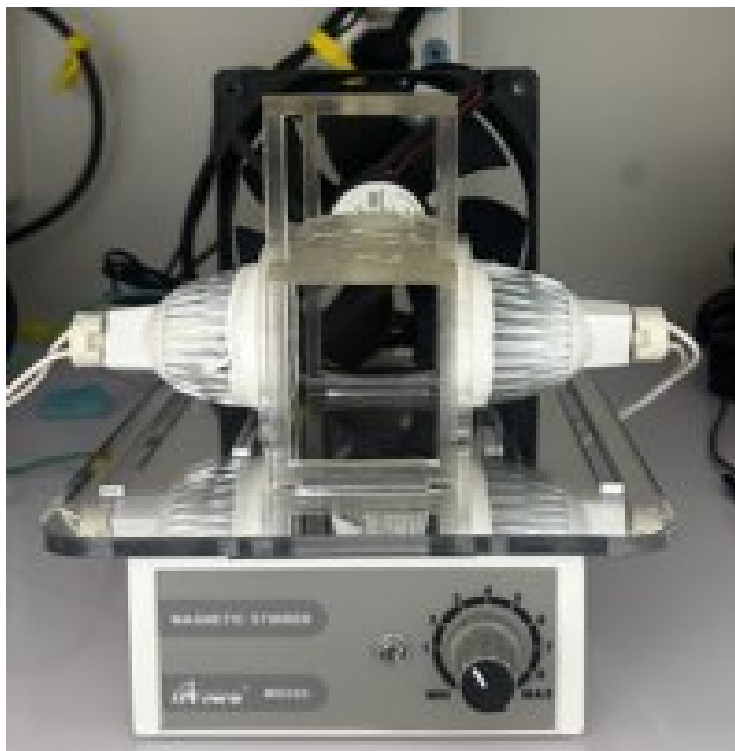


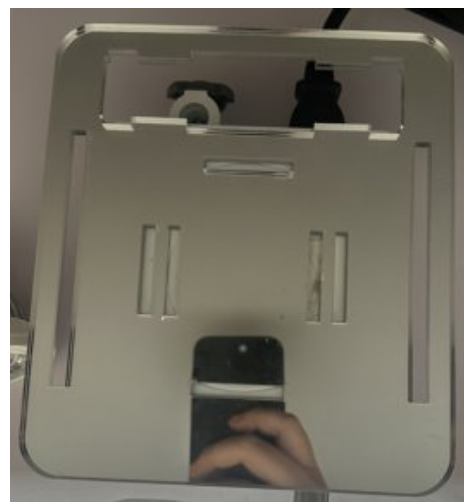
Photo reactor
(Batch)



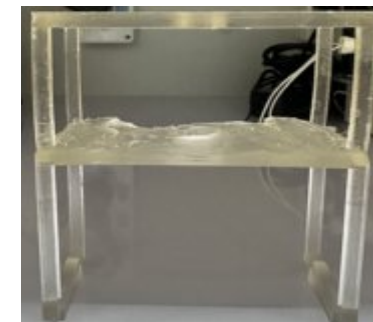
Acrylic LED holder



MR16 LEDs



Acrylic board



Reaction tube holder (1 hole)



Reaction tube holder (2 hole)



Cooling fan

LEDs	Power	Wavelength (Measured)	Light Color	Temperature	Inventory
MR 16	5 W	452 nm	Blue	Fan: r.t. No fan: 40 °C	>20
MR 16	3 W	-	Blue	-	8 ea
MR 16	9 W	455 nm	Blue	-	4 ea
MR 16	5 W	517 nm	Green	-	6 ea
MR16	5 W	-	Purple	-	2 ea
Kessile Tuna blue	40 W	458 nm	Blue	-	1 ea
Kessile 440 nm	40 W	438 nm	Blue	-	1 ea
Kessile 390 nm	40 W	-	Violet	-	1 ea

Tube holder	Inventory
1 hole	7 ea
2 hole	7 ea

사용 가능한 총 Reactor 개수는 **14개** 입니다.